

Ethan Pan

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EDUCATION

2025 - 2029	Computer Science, B.S and Physics, B.S at University of California, Los Angeles	GPA: 3.89
Current Courses	Physics 105B - Analytic Mechanics, Physics 110B - Electricity and Magnetism, Physics 115A - Quantum Mechanics, Computer Science M152A - Introductory Digital Design Laboratory	
2021 - 2024	Saddleback College	
2021 - 2024	Irvine Valley College	
2021 - 2025	Northwood High School	

SKILLS

CAD with Solidworks	C, C++, Python, Java, x86-64 Assembly, NI LABVIEW
Programming languages	Component-level board fault-finding and repair, BGA Rework of smaller chips, PCB trace repair, microsoldering
PCB rework and soldering	DFS/BFS/Floodfill, DSU, MST algorithms, Dijkstra, Dynamic Programming
Algorithmic programming	
Familiarity with common shop tools	
Basic manufacturing skills	

EXPERIENCE

Undergraduate Researcher at S-Pack, UCLA April 2026-Present

Currently working as an undergraduate researcher in Professor Tiwei Wei's lab, supporting PhD student Yunchun Yang's research in 2-phase flow instability for application in server-grade chip cooling. Responsibilities include building both hardware setup and software for data collection and automated control of the sensors, DAQ, power supplies, and high-speed camera, assisting with CAD design of prototypes in Solidworks, manufacturing prototypes using resin 3D printing and laser cutting, and conducting experiments.

USA Computing Olympiad 2021-2024

Competed in the USACO, an algorithmic programming competition, reaching Gold division by 2024.

American Math Competition 2020-2022

Competed in the AMC10, qualifying for the AIME in 2022.

FIRST Robotics Competition 2021 - 2023

Designed various robot subassemblies to solve game problems vital to scoring ability and reliability, turned CAD files into diagrams for machinists with geometric dimensioning annotations, and machined various parts using basic shop tools. Diagnosed failures and jury-rigged emergency repairs under time stress during competitions.

PROJECTS

RX-8 Engine Rebuild June 2025 - present

Acquired a 2008 Mazda RX-8 with a blown rotary engine to repair for personal use. Successfully removed, tore down, and diagnosed out-of-spec side seals using vernier scale caliper, micrometer, dial gauge, and feeler gauges, verifying other parts within margin of reuse. Engine was successfully reassembled according to factory specification.

SAP 8-bit Computer October 2024 - January 2025

Implemented a SAP architecture computer inspired by the specification in Malvino and Brown's *Digital Computer Electronics* using only basic logic gates. Utilized knowledge of boolean algebra, multiplexing for seven-segment display out, reading schematics for component selection, and other computer organization/architecture concepts.

1-D Schrodinger Equation Solver April 2026

Used Python with numpy and matplotlib to simulate the time evolution of a 1-D wavefunction under various boundary conditions, using the Crank-Nicolson Finite Difference method for partial differential equations. Viewable on Github under DILET1/SchrodingerSolver.

Java Game Engine March-August 2024

Created a 2D game engine in Java, using the Processing 4 library for graphics. Capable of generating gameworlds from formatted text file inputs, loading between gameworlds, and persistent game scripting between worlds. Modular scripting

system allows new functions to be added in the future, should the project be re-activated for further development.

Graphics Card Repair

2021 - present

Performed board-level diagnosis and repair of Nvidia and AMD Radeon consumer graphics cards for resale to cryptocurrency miners, primarily GTX 10-series, RX 500-series, RTX 20-series cards. Techniques performed included circuit tracing and diagnosis using a multimeter, oscilloscope, and benchtop voltage injection, as well as trace repair and BGA repair using a hot-air station and IR preheater.